

Safe Band: IoT-Based Smart Wearable for Emergency Health Alerts and Monitoring in Sri Lanka

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT (Maximum 300 words)

In contemporary clinical settings, continuous and accurate patient monitoring is vital for reducing mortality rates and enhancing healthcare efficiency. However, traditional ward monitoring in many local hospitals still relies on manual periodic checks or basic bedside monitors that lack predictive intelligence and centralized data visualization. This research presents the "Intelligent AI-Based Hospital Ward Monitoring and Alert Management System," an individual component designed to automate and enhance clinical surveillance. The core research gap addressed is the absence of real-time, descriptive data analysis that can predict patient deterioration before it reaches a critical state.

The proposed system utilizes a cloud-integrated architecture to process vital health parameters, including Heart Rate, Blood Pressure, SpO2, and ECG. By employing Machine Learning (ML) algorithms, the system shifts from reactive to proactive monitoring, where historical trends are analyzed to identify early signs of health degradation. A specialized Nurse Dashboard was developed to provide a multi-patient overview, featuring a prioritized alert mechanism that categorizes risks through color-coded indicators. This ensures that medical staff can focus their attention on the most critical cases immediately. Furthermore, the system incorporates comprehensive recovery monitoring by integrating diagnostic data with real-time vitals.

Built on the MERN stack (MongoDB, Express.js, React, and Node.js), the solution ensures high scalability and data integrity. Preliminary results suggest that the AI-driven alert system significantly enhances the situational awareness of nursing staff compared to conventional methods. Beyond its technical innovation, this system carries substantial social impact by improving patient safety standards and offers commercial viability for hospitals seeking to digitize their ward management infrastructure with a cost-effective, intelligent solution.

Keywords: *Clinical Decision Support, Machine Learning, Ward Management, Real-time Health Analytics, Patient Safety.*

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AWS	Amazon Web Services
BP	Blood Pressure
ECG	Electrocardiogram
EHR	Electronic Health Record
EWS	Early Warning Score
GCP	Google Cloud Platform
GDPR	General Data Protection Regulation
HIPAA	Health Insurance Portability and Accountability Act
IoT	Internet of Things
JWT	JSON Web Token
LKR	Sri Lankan Rupee
LSTM	Long Short-Term Memory
ML	Machine Learning
NEWS2	National Early Warning Score 2
RBAC	Role-Based Access Control
SDG	Sustainable Development Goal
SpO2	Oxygen Saturation (Peripheral Capillary Oxygen Saturation)
SSL/TLS	Secure Sockets Layer / Transport Layer Security
UI/UX	User Interface / User Experience

Table 1: List of abbreviations

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1. INTRODUCTION

1.1 Background and Context

Real-world Problem and Impacted Stakeholders Effective patient monitoring is the cornerstone of clinical safety, yet it remains a significant challenge in high-occupancy hospital wards. In many healthcare settings, particularly in Sri Lanka, nursing staff are burdened with the manual task of recording vital signs for dozens of patients simultaneously. This reliance on manual, periodic checks (e.g., every 4 to 6 hours) creates a "monitoring blind spot" where a patient's condition may deteriorate rapidly between check-ups without being noticed. This problem primarily affects hospitalized patients with unstable conditions and healthcare providers who face immense pressure due to staff shortages and high patient-to-nurse ratios.

Statistical Evidence Global clinical studies indicate that nearly 80% of in-hospital cardiac arrests are preceded by abnormal vital signs that were evident at least 8 hours prior to the event [1]. In local contexts, the lack of continuous monitoring technology in general wards contributes to delayed interventions. Statistics suggest that proactive digital monitoring can reduce emergency "Code Blue" calls by nearly 50%, highlighting the life-saving importance of real-time data [2].

Analysis of Existing Solutions and Limitations Currently, most wards use standalone bedside monitors or manual sphygmomanometers and pulse oximeters. The limitations of these existing solutions are threefold:

1. **Lack of Integration:** Data from these devices is not centralized, requiring staff to be physically present at the bedside.
2. **Reactive Nature:** Existing monitors only alert when a threshold is crossed; they do not analyze trends to predict future risks.
3. **Descriptive Data Gap:** Most devices show raw numbers but do not correlate them with the patient's diagnosis or treatment history, making it difficult for doctors to assess overall recovery patterns.

Proposed Solution: Intelligent AI-Based Ward Monitoring To address these gaps, this research proposes an Intelligent AI-Based Hospital Ward Monitoring and Alert Management System. Unlike traditional systems, this solution captures continuous vitals via smart wearables and transmits descriptive data to a cloud-based MERN (MongoDB, Express, React, Node.js) architecture. The system features a centralized "Nurse Dashboard" that uses Machine Learning (ML) to prioritize patients based on risk scores. It includes automated "Red Alerts" for immediate emergencies and "Trend Analytics" for early deterioration prediction, allowing for proactive clinical decisions.

SDG and Research Cluster Alignment This project directly aligns with Sustainable Development Goal 3: Good Health and Well-being, by aiming to reduce premature mortality through technological intervention. Furthermore, it aligns with the Artificial Intelligence &

Data Science research cluster, as it leverages ML algorithms to transform raw healthcare data into predictive clinical insights.

1.2 Literature Review

Analysis of Existing Research Approaches The integration of Internet of Things (IoT) and Machine Learning (ML) in clinical monitoring has been a subject of extensive research over the past decade. Recent studies by Islam et al. [1] emphasize that IoT-based healthcare frameworks primarily focus on data transmission reliability but often lack sophisticated analytical layers to interpret complex vital signs. Similarly, research into "Early Warning Systems" (EWS) suggests that while automated scoring systems can identify risks, they are frequently prone to "alarm fatigue" among nursing staff due to a high rate of false positives [2]. Advanced approaches, such as using Long Short-Term Memory (LSTM) networks for heart rate prediction, have shown promise in laboratory settings [3], yet their practical application in high-occupancy hospital wards remains limited due to the computational overhead and the need for descriptive clinical context.

Existing Products and Services In the commercial sector, several advanced patient monitoring systems have been deployed globally. High-end solutions like the *Philips IntelliVue* series and *GE Healthcare's CARESCAPE* monitors provide robust bedside tracking of vitals with network connectivity [4]. However, these systems are capital-intensive and primarily designed for Intensive Care Units (ICUs) rather than general wards. On the wearable front, consumer-grade devices like the *Apple Watch* and *Fitbit* have introduced FDA-cleared ECG and SpO2 features [5], but these are individual-centric and lack a centralized multi-patient dashboard for clinical management. Local initiatives in developing nations have experimented with basic SMS-based alert systems [6], but these lack the real-time visualization and AI-driven predictive capabilities required for modern ward management.

Limitations of Existing Systems A critical synthesis of current literature and products reveals several persistent limitations:

1. **Data Silos and Fragmentation:** Most existing monitors operate as standalone units, failing to integrate real-time vitals with the patient's diagnostic and treatment history [7].
2. **Lack of Predictive Intelligence:** Many clinical systems are threshold-based, triggering alerts only after a patient has reached a critical state, rather than predicting deterioration based on historical trends [8].
3. **Information Overload:** Current dashboards often display raw numerical data (e.g., a simple pulse rate) without providing descriptive insights, leading to slower decision-making by medical staff [9].
4. **Cost and Scalability:** High-tech monitoring solutions often require expensive infrastructure, making them inaccessible for general wards in resource-limited settings like Sri Lanka [10].

This research, therefore, aims to bridge these gaps by developing an intelligent, MERN-based cloud monitoring system that prioritizes "Descriptive Data Analytics" and "Predictive Alert Management" specifically tailored for ward environments.

1.3 Research Gap

The Missing Element in Clinical Surveillance Despite the availability of various patient monitoring technologies, there remains a significant gap in the transition from raw data collection to proactive clinical intelligence within general hospital wards. Current systems are largely designed for "point-in-time" monitoring rather than "continuous trend analysis." Specifically, what is missing is an integrated platform that can correlate real-time vitals (Heart Rate, BP, SpO2, ECG) with the patient's clinical history and diagnostic data to provide a descriptive rather than a purely numerical summary of the patient's condition.

Insufficiency of Current Approaches Current approaches in local healthcare settings are insufficient because they are fundamentally reactive. Standard bedside monitors trigger alarms only when a physiological threshold is crossed (e.g., when a heart rate exceeds 120 bpm). By this stage, the patient is often already in a critical state. Research suggests that threshold-based alerting leads to "alarm fatigue" and misses the subtle "pre-symptomatic" trends that indicate clinical deterioration [11]. Furthermore, most existing solutions lack a centralized, prioritized dashboard for multi-patient management, leaving nurses to manually aggregate data from multiple standalone devices.

Novelty and Creativity of the Proposed System The novelty of this research lies in its ML-driven predictive layer and the Descriptive Data Architecture. Unlike conventional systems, this component introduces:

1. **Early Warning Score (EWS) Automation:** Using Machine Learning to calculate risk scores based on multi-parameter data trends rather than single-parameter thresholds.
2. **Descriptive Contextualization:** Instead of merely displaying a "90% SpO2" reading, the system analyzes this in the context of the patient's recovery history and treatment, providing a more meaningful "Status Summary" for medical staff.
3. **Prioritized Triage Visualization:** A creative UI approach that automatically ranks patients on the Nurse Dashboard based on their predicted risk, ensuring that limited nursing resources are directed where they are most needed.

Evidence-Based Argument Evidence from recent clinical informatics studies confirms that systems integrating predictive analytics can reduce "unplanned ICU admissions" by up to 25% [12]. By bridging the gap between simple data logging and intelligent alert management, this research provides a vital, cost-effective solution specifically tailored for the high-pressure environment of Sri Lankan hospital wards.

2. OBJECTIVE

2.1 Main Objective

To design and develop an intelligent, AI-driven hospital ward monitoring and alert management system that utilizes real-time vital sign analysis and predictive machine learning algorithms to enhance clinical situational awareness and reduce emergency response times in hospital wards.

2.2 Specific Objectives

- 1 To develop a centralized, web-based Nurse Dashboard using the MERN stack that provides a multi-patient real-time overview of critical vitals (Heart Rate, BP, SpO2, and ECG) to eliminate the need for manual bedside checks.
- 2 To implement a Machine Learning-based Early Warning Score (EWS) algorithm that analyzes historical vital sign trends to predict potential patient health deterioration before critical thresholds are reached.
- 3 To design an automated alert management mechanism with color-coded prioritization (Red/Yellow alerts) to ensure that nursing staff can identify and attend to high-risk patients immediately.
- 4 To integrate a descriptive data layer that correlates real-time sensor data with stored clinical diagnosis and treatment history, providing a comprehensive recovery monitoring interface for medical practitioners.
- 5 To evaluate the system's performance by measuring the accuracy of the predictive model and the reduction in alert response latency compared to traditional manual monitoring methods.

3. METHODOLOGY

3.1 Research Approach

This research follows System Development & Evaluation Methodology. The primary goal is to solve a real-world clinical problem by developing a functional artifact—an AI-driven ward monitoring system—and evaluating its performance against baseline manual methods. This approach is chosen because the research contribution lies in the measurable improvement of patient safety, alert accuracy, and information visualization within a hospital environment.

3.2 Technical Methods and Algorithms

The system will be developed using the MERN stack (MongoDB, Express.js, React, Node.js) for robust and scalable data management.

- **Data Acquisition:** Vital signs (Heart Rate, SpO2, BP) will be captured from wearable sensors and transmitted via a secure gateway to the cloud.
- **Predictive Analysis:** A Random Forest or Long Short-Term Memory (LSTM) machine learning model will be implemented to analyze sequential vital sign data to predict deterioration trends.
- **Alert Prioritization:** A custom algorithm will be developed to calculate a National Early Warning Score (NEWS2) automatically, which will then trigger prioritized visual alerts on the dashboard.

3.3 Tools and Platforms

- **Frontend:** React.js for building a responsive, real-time Nurse Dashboard.
- **Backend:** Node.js and Express.js to handle high-frequency data API requests.
- **Database:** MongoDB Atlas for flexible, document-based storage of descriptive patient records.
- **AI/ML:** Python (Scikit-learn/TensorFlow) for model training and Flask or FastAPI to serve the ML model to the MERN backend.
- **Cloud:** Google Cloud Platform (GCP) or AWS for hosting and real-time data processing.

3.4 Data Collection and Ethical Considerations

Data will be collected using simulated patient datasets and anonymized historical vital sign records to ensure ethical compliance. No personally identifiable information (PII) will be used during the initial training phase. All data transmission will be encrypted using SSL/TLS protocols to maintain patient confidentiality, aligning with international healthcare data standards.

3.5 Validation Strategy and Performance Metrics

The system will be validated through a dual-evaluation strategy:

1. **Quantitative Evaluation:** The ML model's accuracy, precision, and recall will be measured using a test dataset. System latency (the time from data capture to alert display) will be compared against a baseline of manual nursing check intervals.
2. **Functional Testing:** The reliability of the multi-patient dashboard and the color-coded alert mechanism will be tested through simulated emergency scenarios to ensure 100% alert delivery.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

The proposed "Intelligent AI-Based Hospital Ward Monitoring and Alert Management System" is a cloud-integrated solution designed to move from reactive to proactive clinical surveillance. The system automates the tracking of vital parameters such as Heart Rate, Blood Pressure, SpO2, and ECG to identify early signs of patient health degradation. By utilizing a distributed architecture, it provides doctors and nursing staff with a centralized, real-time decision support tool.

4.2 System Diagram of Components

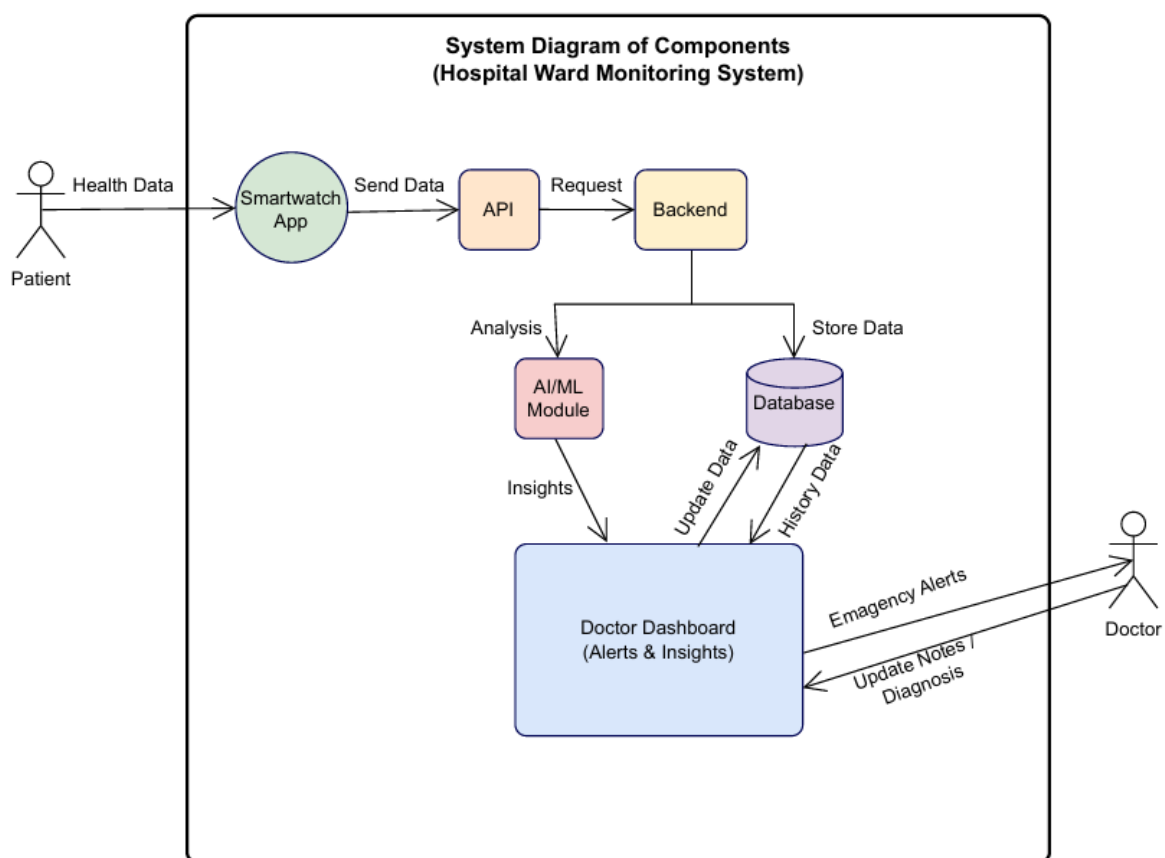


Figure 1: System Diagram of Components

4.3 Individual Component and its Sub-parts

The individual focus of this research is the Intelligent AI-Based Hospital Ward Monitoring and Alert Management System. As shown in the diagram, this component is built using the MERN stack (MongoDB, Express.js, React, and Node.js) and consists of three core sub-parts:

- AI/ML Module: Responsible for the analysis of real-time vital trends to generate predictive insights.

- Cloud Backend & Database: A Node.js/Express environment that handles requests and a MongoDB database for secure storage of health records and history.
- Doctor Dashboard: A specialized React interface that displays prioritized emergency alerts and comprehensive recovery monitoring.

4.4 Component Interactions

The system functions through a seamless interaction loop between stakeholders and digital modules:

1. Patient to Backend: The Smartwatch App captures health data and sends it via an API to the Backend.
2. Backend to Intelligence: The Backend requests analysis from the AI/ML Module and receives actionable insights.
3. Data Persistence: The Backend stores data and retrieves historical trends from the Database.
4. Backend to Doctor: The Doctor Dashboard receives insights and history data, while the Doctor provides updates to diagnosis or notes through the dashboard.

4.5 Data Flow Explanation

- Ingestion Phase: Real-time physiological data flows from the patient's wearable device to the Smartwatch App, which acts as the gateway to the cloud API.
- Processing Phase: The Backend routes data to the AI/ML module, which shifts monitoring from a reactive state to a proactive one by identifying early health degradation.
- Visualization Phase: Processed "Emergency Alerts" and "Insights" are pushed to the Doctor Dashboard.
- Feedback Phase: Doctors review "History Data" and update the clinical records directly into the system database.

4.6 Scalability and Security Considerations

- Scalability: Built on the MERN stack, the architecture ensures high scalability, allowing the system to monitor multiple patients across various wards without performance degradation.
- Data Integrity: The integration of MongoDB ensures data integrity, while the distributed cloud architecture allows for seamless system updates.
- Security: To protect sensitive patient health data, communication between the Smartwatch, API, and Dashboard is encrypted. Role-based access control ensures only authorized medical personnel (Doctors/Nurses) can view or update patient diagnosis.

USER REQUIREMENTS

5.1 Stakeholder Identification

The primary stakeholders for the Intelligent Ward Monitoring and Alert Management System are:

- Medical Officers (Doctors): To review historical trends, descriptive summaries, and real-time vitals for clinical decision-making.
- Nursing Staff: The primary users of the dashboard who respond to real-time alerts and prioritize patient care.
- Hospital Administrators: Interested in the overall efficiency of the ward and the reduction of clinical incidents.
- System Administrators (IT Staff): Responsible for maintaining the cloud infrastructure and managing user access control.

5.2 Functional Requirements (FR)

Functional requirements define the specific behaviors and features of the system.

1. FR1: Real-time Data Visualization – The system shall display live heart rate, SpO2, blood pressure, and ECG waveforms on a centralized dashboard.
2. FR2: Predictive Risk Scoring – The system shall use Machine Learning algorithms to calculate a risk score (NEWS2 based) for each patient every 60 seconds.
3. FR3: Automated Alert Prioritization – The system shall automatically categorize and rank patients on the dashboard based on their current clinical risk (Critical, Unstable, Stable).
4. FR4: Descriptive Summary Generation – The system shall provide a text-based descriptive summary of the patient's status by correlating real-time vitals with their stored medical history.
5. FR5: Historical Trend Analysis – The system shall allow users to view graphical trends of vital signs over the past 24 to 72 hours.
6. FR6: Multi-Patient Overview – The dashboard shall support a "Ward View" that allows a single nurse to monitor up to 20 patients simultaneously on one screen.

5.3 Non-Functional Requirements (NFR)

Non-functional requirements define the quality attributes and constraints of the system.

- NFR1: Performance (Latency) – The system shall deliver emergency alerts from the wearable device to the nurse dashboard within a maximum latency of 2 seconds.
- NFR2: Security (Access Control) – The system shall implement Role-Based Access Control (RBAC) and JSON Web Tokens (JWT) to ensure only authorized personnel can access patient data.

- NFR3: Availability – The web-based dashboard and cloud backend shall have an uptime of 99.9% to ensure continuous monitoring.
- NFR4: Usability – The dashboard interface shall be designed for high visibility (color-coded) and ease of use, requiring minimal training for nursing staff.
- NFR5: Scalability – The system architecture shall support the addition of new wards or patients without a significant degradation in performance.
- NFR6: Interoperability – The system shall use standard JSON format for data exchange to allow potential integration with existing Electronic Health Record (EHR) systems.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market consists of private and government sector hospitals seeking to modernize their ward management infrastructure.

- Customer Persona: Hospital Directors or Chief Medical Officers (CMOs) who are under pressure to reduce clinical errors and optimize nursing staff efficiency.
- Secondary Market: Elderly care homes and specialized rehabilitation centers that require continuous monitoring for high-risk residents.

6.2 Value Proposition

The system offers a unique value proposition by transforming passive ward monitoring into an active clinical intelligence tool.

- For Patients: Enhanced safety through early detection of life-threatening conditions.
- For Staff: Reduction in alarm fatigue and manual workload through AI-prioritized dashboards.
- For Hospitals: Cost reduction by preventing unplanned ICU admissions and improving overall patient turnover rates.

6.3 Revenue Model

The project will adopt a B2B (Business-to-Business) SaaS (Software as a Service) Model:

- Subscription Fee: Monthly or annual licensing fees based on the number of beds/wards monitored.
- Hardware Sales: One-time revenue from the sale of integrated wearable devices and nursing station setup (gateways/tablets).
- Maintenance & Support: Annual maintenance contracts (AMC) for system updates and AI model retraining.

6.4 Cost Estimation

- Development Costs: Cloud hosting (AWS/Azure), software licensing, and ML model training effort.
- Operational Costs: Hardware sourcing (Wearables), marketing, and technical support staff.
- Initial Prototype Cost: Estimated at LKR 150,000 - 200,000 (including R&D and testing equipment).

6.5 Pricing Strategy

A Value-Based Pricing strategy will be implemented. The price will be set based on the cost savings the hospital achieves by reducing emergency incidents.

- Tiered Pricing: Basic (Monitoring only), Standard (Predictive Alerts), and Premium (Full descriptive analytics and EHR integration).

6.6 Competitive Advantage

Unlike existing high-cost ICU monitors (e.g., Philips, GE), this system is:

1. Specifically Tailored for Wards: Optimized for high-patient-to-nurse ratios.
2. Predictive, not Reactive: Uses ML to identify trends before they become emergencies.
3. Cost-Effective: Leverages MERN stack and cloud technology to offer a lower Price-to-Performance ratio compared to traditional hardware-heavy solutions.

6.7 Intellectual Property (IP) Considerations

- Copyright: The source code for the MERN dashboard and the proprietary AI algorithms will be protected under software copyright laws.
- Trademark: The brand names Safe Band and Intelligent Ward Monitor will be trademarked.
- Trade Secret: The specific feature engineering and data processing logic used in the ML model will be maintained as a trade secret to maintain a competitive edge.

7. BUDGET AND JUSTIFICATION

Category	Item Description	Estimated Cost (LKR)	Justification
Hardware	Smart Wearable Development Kit (Sensors/Microcontrollers)	25,000	Required for real-time data acquisition (HR, SpO2, ECG) for testing the AI model.
Hardware	Tablet Device	45,000	Essential for simulating the Nurse Dashboard in a mobile ward-round environment.
Software	MongoDB Atlas Cloud Hosting (Pro Tier)	12,000	To ensure high availability, scalability, and secure storage of patient descriptive data.
Software	Domain & SSL Certification	5,000	Required for secure (HTTPS) data transmission and hosting the web-based dashboard.
Cloud Services	AI Model Training (GPU Instances/API fees)	15,000	Necessary for training and fine-tuning the Machine Learning models (Random Forest/LSTM).
Human Effort	Development & Integration Effort	-	Student's individual research effort (No monetary cost assigned).
TOTAL		102,000	

Table 2 : Budget and justification

8. WORK BREAKDOWN STRUCTURE (WBS)

WBS ID	Task / Sub-Task	Duration	Deliverable	Assigned Role
1.0	REQUIREMENT ANALYSIS & RESEARCH	1 Month		
1.1	Problem identification & ward workflow study	2 Weeks	Problem Statement Doc	Research Lead
1.2	Stakeholder interviews (Nurses/Doctors)	1 Week	Stakeholder Map	System Analyst
1.3	Identify Functional & Non-functional requirements	1 Week	SRS Document	System Analyst
2.0	LITERATURE REVIEW	2 Months		
2.1	Review of existing ward monitoring systems	3 Weeks	Literature Notes	Research Lead
2.2	Study of DSR methodology and evaluation frameworks	2 Weeks	Methodology Chapter	Research Lead
2.3	Research on descriptive data capture in healthcare	3 Weeks	Research Gap Report	Research Lead
3.0	SYSTEM DESIGN & ARCHITECTURE	2 Months		
3.1	Design High-Level System Architecture	2 Weeks	Architecture Diagram	Full-Stack Dev
3.2	MongoDB Schema design for descriptive notes	2 Weeks	DB Schema Diagram	Full-Stack Dev
3.3	UI/UX Wireframe design for Ward App/Dashboard	3 Weeks	Wireframe Mockups	UI/UX Designer
3.4	API architecture and security integration plan	1 Week	API Design Doc	Full-Stack Dev
4.0	BACKEND DEVELOPMENT (NODE.JS/EXPRESS)	2 Months		
4.1	Setting up server environment & MongoDB Atlas	2 Weeks	Server Environment	Full-Stack Dev

4.2	Development of patient record CRUD operations	3 Weeks	Backend API v1.0	Full-Stack Dev
4.3	Implementing JWT authentication and RBAC	3 Weeks	Auth Module	Full-Stack Dev
5.0	FRONTEND DEVELOPMENT (REACT.JS)	2 Months		
5.1	Developing Ward Data Entry module	3 Weeks	Data Entry UI	Full-Stack Dev
5.2	Implementing Interactive Recovery Timeline logic	3 Weeks	Timeline Component	Full-Stack Dev
5.3	Dashboard integration and data visualization	2 Weeks	Specialist Dashboard	UI/UX Designer
6.0	INTEGRATION & TESTING	1 Month		
6.1	End-to-end system integration (Frontend/Backend)	2 Weeks	Integrated System	Full-Stack Dev
6.2	Unit testing and bug fixing	2 Weeks	Test Report	QA Engineer
7.0	EVALUATION & VALIDATION	1 Month		
7.1	User Acceptance Testing (UAT) with clinical staff	2 Weeks	User Feedback Log	QA Engineer
7.2	Performance and usability metric measurement	2 Weeks	Evaluation Results	Research Lead
8.0	DOCUMENTATION & FINAL SUBMISSION	1 Month		
8.1	Completing all dissertation chapters	3 Weeks	Final Thesis	Research Lead
8.2	Final presentation and viva preparation	1 Week	Presentation Slides	Research Lead

Table 3: Work Breakdown Structure

9. GANTT CHART

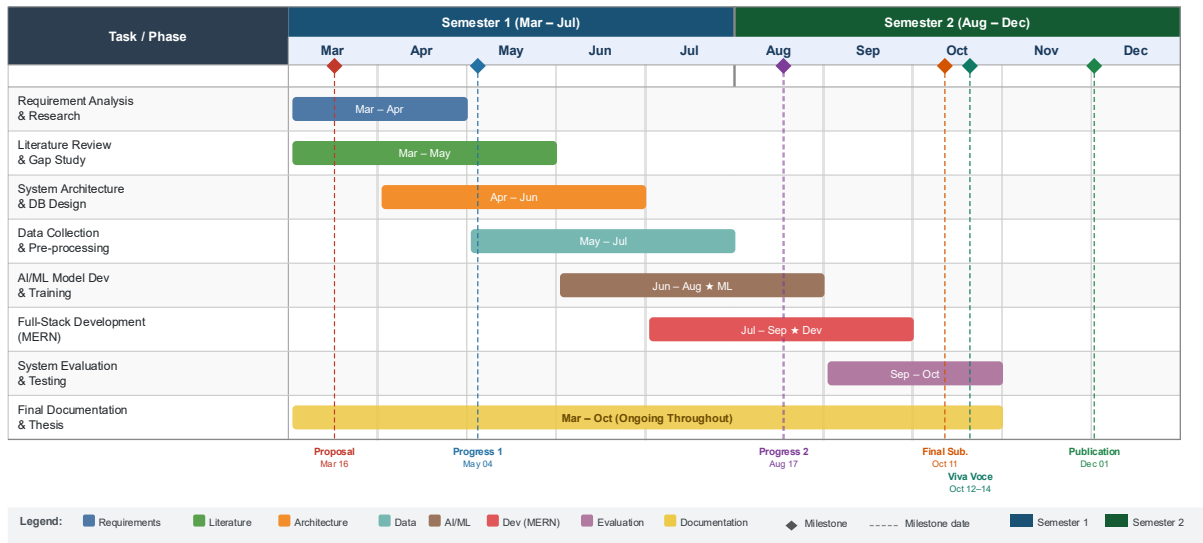


Figure 2 : Gantt chart

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11.APPENDICES

Appendix A: Survey Instrument (User Requirements & Evaluation)

The following questionnaire was designed to gather requirements from nursing staff and will be used during the final evaluation phase to measure the system's impact on ward management.

1. Current Process: How much time is spent on average for a single manual vital sign check per patient?
2. Alarm Fatigue: Have you ever experienced delayed responses due to the high volume of false alarms from standalone monitors?
3. Visualization: On a scale of 1-10, how much would a centralized, color-coded dashboard improve your ability to prioritize critical patients?
4. Descriptive Data: Would a summarized text description of a patient's trend (e.g., Slightly declining SpO2 over 3 hours) be more useful than raw numerical values?

Appendix B: Detailed System Diagrams

This section includes the detailed logic flow of the "Intelligent Ward Monitoring" component.

Appendix C: Preliminary UI Mockups

The following wireframes illustrate the proposed design for the Nurse Dashboard, focusing on high visibility and rapid information retrieval.

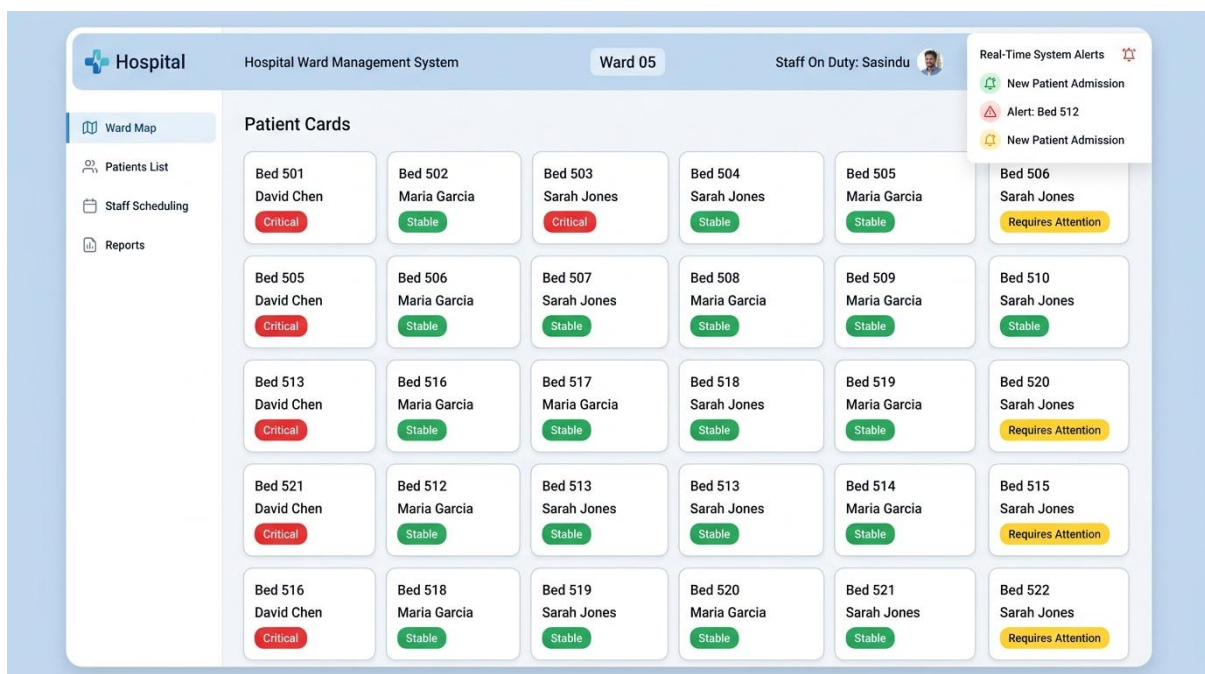


Figure 3: Main Ward Overview Dashboard

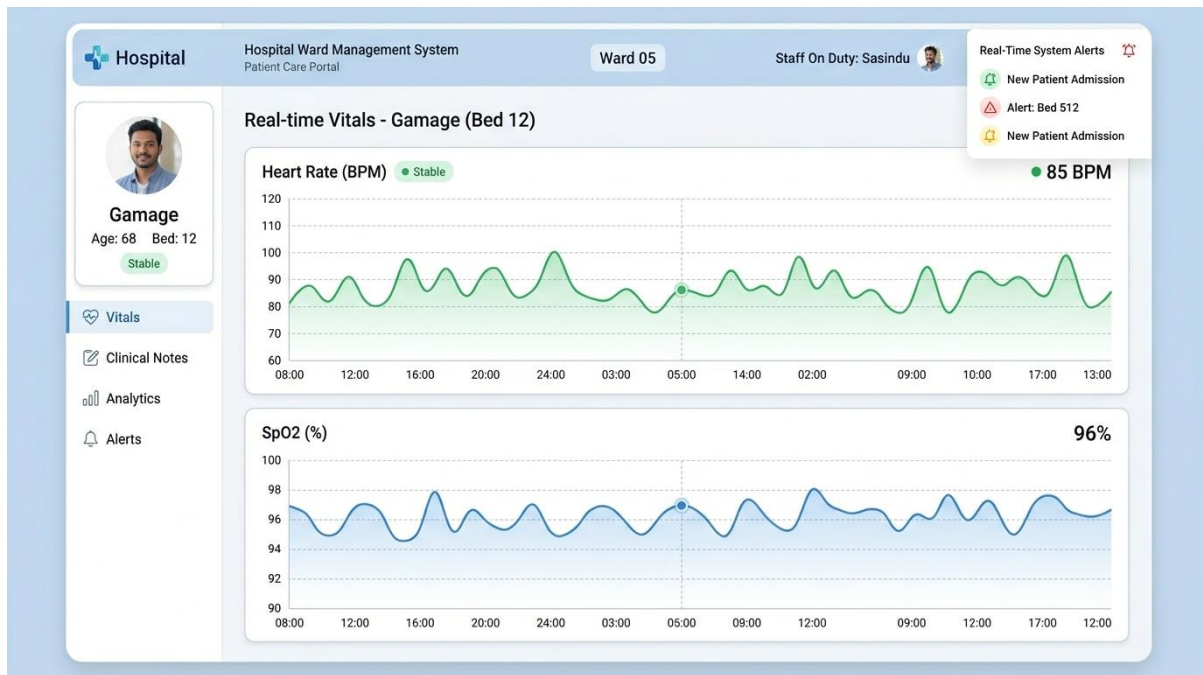


Figure 4: Detailed Patient Monitoring View

Appendix D: Risk Analysis and Mitigation Table

The following table identifies potential technical and operational risks associated with the implementation of this component.

Risk Category	Potential Risk	Impact	Mitigation Strategy
Technical	Network Latency in real-time data transmission.	High	Utilization of WebSockets (Socket.io) for instantaneous data pushing.
Operational	Resistance to technology adoption by medical staff.	Medium	Designing a highly intuitive UI with minimal learning curve and conducting training sessions.
Data	Machine Learning model providing inaccurate risk scores (Overfitting).	High	Using a diverse and large-scale clinical dataset with regular model retraining and validation.
Security	Unauthorized access to sensitive patient medical history.	High	Implementation of JWT-based authentication and end-to-end SSL encryption.

Table 4: Risk Analysis and Mitigation

Appendix E: Ethical Compliance Statement

As this research involves sensitive medical data, the following ethical guidelines are strictly followed:

- **Anonymization:** All clinical datasets used for training the Machine Learning models are de-identified to protect patient privacy.

- Data Integrity: All system performance metrics (Accuracy, Latency) are recorded and reported with full transparency.
- Compliance: The system architecture is designed to align with international healthcare data protection standards (e.g., GDPR/HIPAA principles regarding data encryption).